

Introductory Concepts of *Probability*

Formula Reference Sheet & Graded Problem Set with Full Solutions · Dr. Adebayo F. A.

● Chapter 1 — Set Theory ● Chapter 2 — Probability ● Chapter 3 — Conditional Probability & Bayes ● Exam Traps

CHAPTER 01 Set Theory

DEFINITION

Set Notation

$x \in A$ — x is an element of A

$x \notin A$ — x is not an element of A

$A \subset B$ — A is a subset of B

Two sets $A = B$ iff $A \subset B$ and $B \subset A$.

DEFINITION

Power Set

$\mathcal{P}(A)$ = set of all subsets of A

If $|A| = n$, then $|\mathcal{P}(A)| = 2^n$.

e.g.

$A = \{1, 2, 3\} \Rightarrow \mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, A\}$

OPERATION

Intersection

$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

Disjoint: $A \cap B = \emptyset$

OPERATION

Union

$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$

"or" is inclusive.

OPERATION

Complement

$\bar{A} = A^c = \{x \mid x \in S, x \notin A\}$

$A \cup \bar{A} = S, A \cap \bar{A} = \emptyset$

OPERATION

Set Difference

$A - B = \{x \mid x \in A, x \notin B\} = A \cap \bar{B}$

$\bar{A} = S - A$

THEOREM

De Morgan's Laws

$$\overline{A \cup B} = \bar{A} \cap \bar{B} \quad \overline{A \cap B} = \bar{A} \cup \bar{B}$$

Generalised: $\overline{\bigcup_{i=1}^n A_i} = \bigcap_{i=1}^n \bar{A}_i$ and $\overline{\bigcap_{i=1}^n A_i} = \bigcup_{i=1}^n \bar{A}_i$

THEOREM

Distributive Laws

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

PROPERTY	STATEMENT
Commutativity	$A \cup B = B \cup A, A \cap B = B \cap A$
Associativity	$(A \cup B) \cup C = A \cup (B \cup C)$
Identity	$A \cup \emptyset = A, A \cap S = A$
Annihilation	$A \cap \emptyset = \emptyset, A \cup S = S$
Idempotency	$A \cup A = A \cap A = A$
Monotone subset	$A \cap B \subset A \subset A \cup B$
Useful identity	$(A - B) - C = A - (B \cup C)$

CHAPTER 02 Probability Concepts

DEFINITION

Sample Space & Events

S = set of all possible outcomes

Event: any subset $A \subseteq S$. Elementary event: $|\{a\}| = 1$.

Mutually exclusive: $A \cap B = \emptyset$.

FREQUENCY INTERPRETATION

Empirical Probability

$$P(A) = \lim_{n \rightarrow \infty} \frac{r}{n}$$

r = number of times A occurs in n trials.

AXIOMS (KOLMOGOROV)

Axioms of Probability**AI.** $0 \leq P(A) \leq 1$ for all $A \subseteq S$ **AII.** $P(S) = 1$ **AIII.** $A \cap B = \emptyset \Rightarrow P(A \cup B) = P(A) + P(B)$ General: $P(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n P(A_i)$ when pairwise mutually exclusive

THEOREM 2.2.1

Complement Rule

$$P(\bar{A}) = 1 - P(A)$$

THEOREM 2.2.2

Partition Rule

$$P(A) = P(A \cap B) + P(A \cap \bar{B})$$

Follows from $A = (A \cap B) \cup (A \cap \bar{B})$, which are disjoint.

THEOREM 2.2.3

Addition Rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

THEOREM 2.2.4

Monotonicity

$$A \subset B \Rightarrow P(A) \leq P(B)$$

EQUALLY LIKELY OUTCOMES

Classical Probability Formula

$$P(A) = \frac{n(A)}{N}$$

 $n(A)$ = number of favourable outcomes; $N = n(S)$ = total number of equally likely outcomes.

CHAPTER 2 – COUNTING

Permutations & CombinationsPermutations of n objects: $n!$ Permutations of n taken r at a time: ${}_nP_r = \frac{n!}{(n-r)!}$ Permutations with repetition: n^r Combinations: $\binom{n}{r} = \frac{n!}{(n-r)!r!}$ Multinomial coefficient: $\binom{n}{n_1, n_2, \dots, n_r} = \frac{n!}{n_1!n_2!\dots n_r!}$ $\sum_{i=1}^r n_i = n$. Counts ways to divide n distinct items into r groups of sizes $n_1, \dots, n_r$.

DEFINITION 3.1.1

Conditional Probability

$$P(B | A) = \frac{P(A \cap B)}{P(A)}, \quad P(A) > 0$$

Satisfies all three probability axioms; $P(B | A) + P(\bar{B} | A) = 1$.

MULTIPLICATION RULE

Chain Rule

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1) P(A_2 | A_1) P(A_3 | A_2, A_1) \dots P(A_n | A_{n-1}, \dots, A_1)$$

DEFINITION 3.3.1

Independence

$$A \text{ and } B \text{ independent} \Leftrightarrow P(A \cap B) = P(A) P(B)$$

Equivalently $P(B | A) = P(B)$. If $A \perp B$ then also $A \perp \bar{B}$, $\bar{A} \perp B$, $\bar{A} \perp \bar{B}$.

THEOREM

Law of Total Probability

$$P(B) = \sum_{i=1}^n P(B | A_i) P(A_i)$$

Requires $\{A_i\}$ to be pairwise mutually exclusive and exhaustive: $\bigcup_i A_i = S$.

THEOREM 3.2.1

Bayes' Theorem (General Form)

$$P(A_k | B) = \frac{P(B | A_k) P(A_k)}{\sum_{i=1}^n P(B | A_i) P(A_i)}$$

$P(A_k)$ = prior; $P(B | A_k)$ = likelihood; $P(A_k | B)$ = posterior.

Bayes' Theorem (Two-Event Form)

$$P(A \mid B) = \frac{P(B \mid A) P(A)}{P(B \mid A) P(A) + P(B \mid \bar{A}) P(\bar{A})}$$

FORMULA	EXPRESSION	NOTES
Conditional probability	$P(B \mid A) = \frac{P(A \cap B)}{P(A)}$	—
Complement of conditional	$P(\bar{B} \mid A) = 1 - P(B \mid A)$	—
Partition of A	$P(A) = P(A \cap B) + P(A \cap \bar{B})$	—
Addition rule	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	—
Independence check	$P(A \cap B) \stackrel{?}{=} P(A)P(B)$	also check $P(A \cup B) \leq 1$
Total probability (2 parts)	$P(B) = P(B \mid A)P(A) + P(B \mid \bar{A})P(\bar{A})$	—



Common Exam Traps & Pitfalls

Mistakes that cost marks — know these before you sit the exam.

SET THEORY · TRAP 1



Confusing $\subset$ with $\in$

✗ $D \in C$ when $D = \{2, 4\}$ and $C = \{1, 2, 3, 4\}$

✓ $D \subset C$ — a *set* is a subset, not an element

$\in$ means "is an element of"; the element must be a single object, not a set (unless the set explicitly contains sets as members).

SET THEORY · TRAP 2



Mixing discrete and continuous sets

✗ $C = E$ because both involve 1 to 4

✓ $C = \{1, 2, 3, 4\}$ (integers) $\neq E = [1, 4]$ (all reals)

Always check whether a set is defined by listing integers or by an inequality over the reals before claiming equality.

PROBABILITY · TRAP 3

Forgetting to subtract the intersection

✗ $P(A \cup B) = P(A) + P(B)$

✓ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

The simple sum double-counts outcomes in $A \cap B$. Only omit the subtraction when events are mutually exclusive ($P(A \cap B) = 0$).

PROBABILITY · TRAP 4

Checking impossibility constraints

✗ Accepting $P(A \cap B) > P(A)$ or
 $P(A \cup B) < \max(P(A), P(B))$

✓ Both violate monotonicity: $A \cap B \subseteq A$
 forces $P(A \cap B) \leq P(A)$

Before computing, verify:

$P(A \cap B) \leq \min(P(A), P(B))$ and

$P(A \cup B) \geq \max(P(A), P(B))$.

CONDITIONAL · TRAP 5

Reversing $P(A | B)$ and $P(B | A)$

✗

$P(\text{Machine A} | \text{defective}) = P(\text{defective} | \text{Machine A})$

✓ Apply Bayes' Theorem — the two are generally unequal

This is the "prosecutor's fallacy." Always identify which event is given and which is being sought.

CONDITIONAL · TRAP 6

Independence $\neq$ Mutual Exclusivity

✗ If $P(A \cap B) = 0$ then A and B are independent

✓ Mutual exclusivity means $P(A \cap B) = 0$;
 independence requires
 $P(A \cap B) = P(A)P(B)$

Two non-trivial mutually exclusive events are dependent (knowing one occurred rules out the other).

BAYES · TRAP 7

Ignoring base rates (prior probability)

✗ A 98% accurate test $\implies$ positive result means 98% chance of disease

✓ With a rare disease (e.g. 3% prevalence), most positives are false positives — use Bayes

Low prior probability can make $P(\text{disease} | +)$ far smaller than the test's sensitivity. Always include $P(C)$ in the denominator via total probability.

COUNTING · TRAP 8

Order matters: permutation vs combination

✗ Using $\binom{n}{r}$ when the arrangement (order) is significant

✓ Use ${}_nP_r = n!/(n-r)!$ when order matters; $\binom{n}{r}$ when only selection matters

Ask: "Does swapping two chosen items give a different outcome?" If yes, use permutations.

Forgetting zero-digit restrictions

✗ Treating each remaining digit position as having 9 choices after fixing the first

✓ If the second position also excludes 0 AND the first digit, count carefully: $10 - 2 = 8$ choices

Work position by position, accounting for all restrictions already applied.

Partition must be exhaustive

✗ Using

$$P(B) = P(B | A_1)P(A_1) + P(B | A_2)P(A_2)$$

when $A_1 \cup A_2 \neq S$

✓ Ensure $\{A_i\}$ covers all of S : $\sum_i P(A_i) = 1$

Missing partitions cause the computed total probability to be less than the true value.



Problem Sheets — Graded Questions

Questions drawn directly from the STA122 Problem Sheets. Difficulty is indicated for each item.

PROBLEM SHEET 1 — SET THEORY

Q1

EASY

Let $U = \{1, 2, 3, \dots, 10\}$. Three subsets are defined as $A = \{1, 2, 3, \dots, 8\}$, $B = \{2, 3, 4\}$ and $C = \{1, 3, 5, 7, 9\}$.

- Find $\bar{A}$, $B \cup \bar{C}$, and $A \cap (\bar{B} \cup C)$.
- Is it true that $\bar{B} \subset A \cap C$?

Q2

MEDIUM

Let $A = \{x \mid 0$

1. $C = E$
2. $D \subset C$
3. $D \in E$
4. $E \not\subset A$

Q3

HARD

Show algebraically that $(A - B) - C = A - (B \cup C)$ using set identities.

Q4

MEDIUM

Let S be the set of all companies on the Botswana Stock Exchange and define: A = mining sector, B = turnover $> P10m$, C = financial year ending in June, D = share price higher than 6 months ago. Describe in words the set $B \cap (C \cup A)$.

PROBLEM SHEET 2 – PROBABILITY FOUNDATIONS

Q5

EASY

Let A and B be events in a sample space with $P(A) = 0.3$, $P(B) = 0.4$ and $P(A \cap B) = 0.2$. Find the probability of each of the following events:

1. Either A or B occurs.
2. A does not occur.
3. Neither A nor B occurs.
4. Exactly one of A or B occurs.

Q6

MEDIUM

Determine which of the following combinations are *impossible* (i.e. violate the axioms of probability):

1. $P(A) = 0.7$, $P(B) = 0.9$, $P(A \cap B) = 0.4$
2. $P(A) = 0.2$, $P(B) = 0.5$, $P(A \cap B) = 0.25$
3. $P(A) = 0.8$, $P(B) = 0.4$, $P(A \cup B) = 0.7$

Q7

HARD

Show that $P(A \cup \bar{B}) = 1 - [P(B) - P(A \cap B)]$.

PROBLEM SHEET 3 – COMBINATORIAL PROBABILITY

Q8

MEDIUM

What is the probability that an 8-digit cellphone number has no repeated digits, given that the number must start with 7 and the second digit cannot be 0?

Q9

HARD

A pack of 52 cards is dealt into four bridge hands of 13 cards each. Find the probability that a given hand contains:

1. 3 spades, 4 diamonds, 1 heart and 5 clubs.
2. Exactly 3 aces and exactly 4 picture cards (J, Q, K).

Q10

MEDIUM

A class of 75 students is divided into tutorial groups of 25, 30 and 20.

1. In how many ways can this be done?
2. Two brothers are in the class. What is the probability they end up in the same tutorial group?

PROBLEM SHEET 4 – CONDITIONAL PROBABILITY & BAYES

Q11

EASY

Suppose events A and B satisfy $P(A) = 0.6$, $P(B) = 0.2$, and $P(A | B) = 0.5$. Find:

1. $P(B | A)$
2. $P(A \cup B)$
3. $P(\bar{A} \cap B)$
4. $P(B | \bar{A})$

Q12

MEDIUM

Two machines produce the same item. Machine A produced 40% of last week's output; Machine B the remaining 60%. Defect rates are 10% for A and 4% for B.

1. What proportion of the entire output was defective?
2. Given that a randomly selected item is defective, what is the probability it came from Machine A?

Q13

HARD

An X-ray test detects a disease present in 3% of the population. The test falsely reports positive for 7% of healthy people and misses 2% of diseased people.

1. What proportion of diseased people are correctly diagnosed?
2. Of those with a positive reaction, what proportion actually have the disease?
3. Of those with a negative reaction, what proportion actually have the disease?
4. What proportion of tests give an incorrect diagnosis?

Q14

MEDIUM

If $P(A) = 0.6$, $P(B | A) = 0.3$ and $P(A \cup B) = 0.72$, determine whether A and B are independent, mutually exclusive, or neither.

Q15

HARD

A student passes Probability with probability 0.8 if she studied and 0.3 if she did not. 60% of the class studied. Given that a randomly chosen student passed, what is the probability she studied?

Answer Sheet — Full Solutions

Complete worked solutions for all fifteen questions above.

SOLUTIONS – PROBLEM SHEET 1

A1

- $\bar{A} = \{9, 10\}$
- $\bar{C} = \{2, 4, 6, 8, 10\}$, so $B \cup \bar{C} = \{2, 3, 4, 6, 8, 10\}$
- $\bar{B} = \{1, 5, 6, 7, 8, 9, 10\}$, $\bar{B} \cup C = \{1, 3, 5, 6, 7, 8, 9, 10\}$, so $A \cap (\bar{B} \cup C) = \{1, 3, 5, 6, 7, 8\}$
- Is $\bar{B} \subset A \cap C$? We have $\bar{B} = \{1, 5, 6, 7, 8, 9, 10\}$ and $A \cap C = \{1, 3, 5, 7\}$. Since $6 \in \bar{B}$ but $6 \notin A \cap C$, the statement is **FALSE**.

A2

- $C = E$? $C = \{1, 2, 3, 4\}$ (integers) while $E = \{x : 1 \leq x \leq 4\}$ (all reals in $[1, 4]$). **FALSE**
- $D \subset C$? $D = \{2, 4\} \subset \{1, 2, 3, 4\} = C$. **TRUE**
- $D \in E$? D is a set, not a real number; it cannot be an element of E . **FALSE**
- $E \not\subset A$? Every $x \in E = [1, 4]$ satisfies $x \in A$. **FALSE**.

A3

$$(A - B) - C = (A \cap \bar{B}) \cap \bar{C} = A \cap (\bar{B} \cap \bar{C}) = A \cap \overline{(B \cup C)} = A - (B \cup C). \quad \square$$

A4

$B \cap (C \cup A)$ is the set of companies whose annual turnover exceeds P10 million *and* whose financial year ends in June *or* which are in the mining sector (or both).

A5

$$1. P(A \cup B) = 0.3 + 0.4 - 0.2 = 0.5$$

$$2. P(\bar{A}) = 1 - 0.3 = 0.7$$

$$3. P(\overline{A \cup B}) = 1 - 0.5 = 0.5$$

$$4. P(\text{exactly one}) = P(A \cup B) - P(A \cap B) = 0.5 - 0.2 = 0.3$$

A6

$$1. P(A \cap B) = 0.4 \leq \min(P(A), P(B)) = 0.7. \checkmark \text{ Possible.}$$

$$2. P(A \cap B) = 0.25 > P(A) = 0.2. \text{ Since } A \cap B \subset A, \text{ we need } P(A \cap B) \leq P(A). \text{ Impossible.}$$

$$3. P(A \cup B) = 0.7. \text{ Check: must have } P(A \cup B) \geq \max(P(A), P(B)) = 0.8. \text{ But } 0.7 < 0.8.$$

Impossible.

A7

From the partition rule, $P(B) = P(B \cap A) + P(B \cap \bar{A})$, so $P(B \cap \bar{A}) = P(B) - P(A \cap B)$.

Then $P(A \cup \bar{B}) = 1 - P(\bar{A} \cap B) = 1 - [P(B) - P(A \cap B)]$. $\square$

A8

First digit is fixed as 7 (1 choice). Second digit: 0 is excluded and 7 already used, leaving $10 - 2 = 8$ choices. Digits 3–8 are chosen from the remaining 8 digits without repetition.

Total (no restrictions beyond start=7, no-repeat): $1 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3$.

Favourable (second digit $\neq 0$): $1 \times 8 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3$.

$$P = \frac{1 \times 8 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3}{1 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3} = \frac{8}{9} \approx$$

0.8889

A9

$$1. \quad P = \frac{\binom{13}{3} \binom{13}{4} \binom{13}{1} \binom{13}{5}}{\binom{52}{13}}$$

$$= \frac{286 \times 715 \times 13 \times 1287}{\binom{52}{13}} \approx 0.00539$$

2. Aces: $\binom{4}{3}$. Picture cards (J,Q,K — 12 total): $\binom{12}{4}$. Remaining 6 from 36 others: $\binom{36}{6}$.

$$P = \frac{\binom{4}{3} \binom{12}{4} \binom{36}{6}}{\binom{52}{13}} \approx$$

0.00412

A10

$$1. \quad \binom{75}{25, 30, 20} = \frac{75!}{25! 30! 20!} \approx 5.99 \times 10^{35}$$

2. Fix both brothers in the same group — three cases:

- Both in group of 25: $\binom{73}{23, 30, 20}$
- Both in group of 30: $\binom{73}{25, 28, 20}$
- Both in group of 20: $\binom{73}{25, 30, 18}$

$$P = \frac{\binom{73}{23, 30, 20} + \binom{73}{25, 28, 20} + \binom{73}{25, 30, 18}}{\binom{75}{25, 30, 20}}$$

Simplifies via the multinomial identity to:

$$P = \frac{25 \cdot 24 + 30 \cdot 29 + 20 \cdot 19}{75 \cdot 74} = \frac{600 + 870 + 380}{5550} = \frac{1850}{5550} =$$

$$\frac{1}{3} \approx 0.3333$$

A11

$$1. P(A \cap B) = P(A | B) P(B) = 0.5 \times 0.2 = 0.10. \text{ Hence } P(B | A) = \frac{0.10}{0.60} = \frac{1}{6} \approx 0.167$$

$$2. P(A \cup B) = 0.6 + 0.2 - 0.1 = 0.7$$

$$3. P(\bar{A} \cap B) = P(B) - P(A \cap B) = 0.2 - 0.1 = 0.1$$

$$4. P(B | \bar{A}) = \frac{P(B \cap \bar{A})}{P(\bar{A})} = \frac{0.1}{0.4} = 0.25$$

A12

1. Let D =defective, M_A =from Machine A. By total probability:

$$P(D) = P(D | M_A)P(M_A) + P(D | M_B)P(M_B) = 0.10(0.4) + 0.04(0.6) = 0.04 + 0.024 =$$

0.064

2. By Bayes' theorem:

$$P(M_A | D) = \frac{0.10 \times 0.4}{0.064} = \frac{0.04}{0.064} =$$

0.625

A13

Let C =has disease (prior $P(C) = 0.03$), $+$ =positive test.

$$1. \text{ Sensitivity} = P(+ | C) = 1 - 0.02 = 0.98$$

$$2. \text{ By Bayes: } P(C | +) = \frac{0.98 \times 0.03}{0.98(0.03) + 0.07(0.97)} = \frac{0.0294}{0.0294 + 0.0679} = \frac{0.0294}{0.0973} \approx 0.302$$

$$3. P(C | -) = \frac{P(- | C)P(C)}{P(-)} = \frac{0.02 \times 0.03}{0.02(0.03) + 0.93(0.97)} = \frac{0.0006}{0.0006 + 0.9021} \approx 0.000665$$

4. Incorrect: false positive + false negative.

$$P(\text{error}) = P(+ | \bar{C})P(\bar{C}) + P(- | C)P(C) = 0.07(0.97) + 0.02(0.03) = 0.0679 + 0.0006 \approx 0.0685$$

A14

$$P(A \cap B) = P(B | A) P(A) = 0.3 \times 0.6 = 0.18.$$

$$\text{Check addition rule: } P(A \cup B) = 0.6 + P(B) - 0.18 = 0.72 \Rightarrow P(B) = 0.30.$$

$$\text{Independence test: } P(A)P(B) = 0.6 \times 0.30 = 0.18 = P(A \cap B). \quad \text{A and B are independent.}$$

(They are not mutually exclusive since $P(A \cap B) = 0.18 \neq 0$.)

A15

Let St =studied, Pa =passed. Given: $P(Pa | St) = 0.8$, $P(Pa | \overline{St}) = 0.3$, $P(St) = 0.6$.

Total probability: $P(Pa) = 0.8(0.6) + 0.3(0.4) = 0.48 + 0.12 = 0.60$.

$$\text{Bayes: } P(St | Pa) = \frac{0.8 \times 0.6}{0.60} = \frac{0.48}{0.60} = 0.8$$